

Moamen Moustafa Basyoni

Senior .NET Developer | Distributed Backend & AI-Integrated Systems

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PROFESSIONAL SUMMARY

Backend engineer with around three years building enterprise .NET systems — designing REST and gRPC APIs from specification, operating resilient distributed services with asynchronous messaging (RabbitMQ) and Temporal Saga orchestration, and modernizing legacy modules onto contemporary .NET (8/9/10) with Clean Architecture and SOLID. Strong in C#, ASP.NET Core, Entity Framework Core over SQL Server and PostgreSQL, OAuth2 / OpenID Connect / JWT, and asynchronous and concurrent programming. Builds AI-integrated backend tooling: internal Model Context Protocol (MCP) servers, automated code-review pipelines, and multi-model LLM orchestration. Runs technical code reviews and mentors engineers.

PROFESSIONAL EXPERIENCE

.NET Backend Engineer — Andalusia Health Business Solution (AHBS)

Dec 2023 – Present · Alexandria, Egypt

- Build enterprise backend systems on modern .NET (8/9/10) for the Andalusia hospital group across Egypt and Saudi Arabia, designing REST and gRPC API contracts before implementation.
- Create resilient distributed services with fault-tolerant asynchronous messaging (RabbitMQ) and long-running distributed transactions via Temporal Saga orchestration, improving 10+ inter-service interaction patterns.
- Integrate legacy systems with modern platforms — migrated modules from .NET Framework 4.5 / ASP.NET Core 2.2 toward modern .NET with Clean Architecture and SOLID across 5 core and 7+ additional modules.
- Design efficient data access with Entity Framework Core and LINQ over SQL Server and PostgreSQL.
- Built a runtime Business Rule Engine for injected business logic, improving rule-execution performance by 60% (measured with BenchmarkDotNet).
- Implemented an identity/authentication server with OpenID Connect and OAuth2 (OpenIddict), JWT access/refresh tokens, and scope- and claim-based authorization.
- Built internal Model Context Protocol (MCP) servers, automated code-review / pull-request analysis tooling, and multi-model LLM orchestration (OpenAI, OpenRouter) with observability (LangSmith, LangFuse), guardrails, token budgeting, and cost optimization.
- Conduct technical code reviews and mentor engineers; led an 8-session onboarding program reaching 25 engineers across 6 teams.

Backend Engineer — Trastain

Feb 2025 – Aug 2025 · Cairo, Egypt (Remote) · Part-time · PropTech marketplace

- Built a PropTech marketplace backend from scratch with booking, policy enforcement, and RAG-based preference matching.
- Integrated Stripe and Paymob payment gateways with webhook processing and HMAC signature verification.
- Automated build and deployment with GitHub Actions CI/CD to Azure Web Apps and Azure Static Web Apps.

Software Engineer — MYM (Make Your Miracle)

Jun 2023 – Oct 2023 · Giza, Egypt · SaaS / e-commerce / booking

- Developed backend systems for e-commerce and booking platforms end-to-end; integrated payment gateways with webhook handling; deployed to Azure App Service.

TECHNICAL SKILLS

- **Languages:** C#, SQL, Python, JavaScript, TypeScript
- **.NET & Backend:** ASP.NET Core, Minimal APIs, ASP.NET Core MVC, .NET 8/9/10, .NET Aspire, Entity Framework Core, LINQ, REST APIs, gRPC, Webhooks
- **Architecture & Practices:** Clean Code, SOLID, Design Patterns, Clean Architecture, Microservices, Service-Oriented Architecture, Saga orchestration, spec-first API design, technical code review
- **Distributed Systems & Messaging:** RabbitMQ, Temporal, Consul (service discovery, health checks, KV), event-driven architecture, asynchronous & concurrent programming (async/await)
- **Databases:** SQL Server, PostgreSQL, Neo4j, Elasticsearch
- **Identity & Security:** OpenID Connect (OIDC), OAuth2, OpenIddict, JSON Web Tokens (JWT), OWASP-aware secure coding, scope- and claim-based authorization
- **Observability:** LangSmith, LangFuse, prompt telemetry, token & cost monitoring
- **Testing & Performance:** pytest, pytest-asyncio, automated testing, BenchmarkDotNet, performance tuning
- **DevOps:** Docker, Azure DevOps (CI/CD), GitHub Actions, Git, Git Flow
- **AI Engineering:** Model Context Protocol (MCP), LangGraph, CrewAI, RAG, GraphRAG, multi-model orchestration (OpenAI, OpenRouter), prompt engineering, guardrails, human-in-the-loop

EDUCATION

B.Sc. Computer Engineering — Behera Higher Institute (BHI), Alexandria, Egypt · 2023 Grade: B+ · GPA 3.2 / 4.0

PUBLICATION & LANGUAGES

- *IoT-Enabled E-Prescription Management and Dispensing Machine Monitoring* — IEEE, 2024

- Arabic (Native) · English (B2, Upper Intermediate)